CEE 260/MIE 273: Probability and Statistics in Civil Engineering

Lecture M4b: Sampling Variability and Central Limit Theorem

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Outline

1. Method of moments
2. CLT
3. Outlook

Sample moments

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- The moments of a random variable are its key descriptors.

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$$s^2 = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \overline{x})^2 \tag{2}$$

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(c) Now, estimate the variance replacing the denominator $(n-1)$ with n in the estimator s^2. What do you notice?

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$\hat{\sigma}^2$ underestimates σ^2 by 0.031 squared units.

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- The product of large number of random components approaches the lognormal distribution

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Sample proportion and the CLT

If the observations in a given sample are a Bernoulli sequence with a constant proportion (or probability) p, then if n is large, the sample proportion $\hat{p}$ follows a normal distribution (according to the CLT):

$$\hat{p} \sim \mathcal{N}(\mu_{\hat{p}}, SE_{\hat{p}}^2) = \mathcal{N}\left(p, \frac{p(1-p)}{n}\right) \tag{11}$$

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$$\text{Sampling error/standard error of } \hat{p}\text{: } SE_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}} \approx \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

One rule of thumb for determining whether n is large enough is to check that both np and $n(1-p)$ are ≥ 10 (also known as the success-failure condition).

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Example 2: Solar energy expansion (CLT)

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(a) According to the CLT, what is the distribution of $\hat{p}$?

(b) According to the CLT, what are $\mu_{\hat{p}}$ and $SE_{\hat{p}}$, respectively?

CLT application: sample proportion (cont.)

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(a) First, we note that the response of each American adult in the entire population is part of a Bernoulli sequence with $p = 0.88$. According to the CLT, the distribution of $\hat{p}$ (sample proportion) is normal/Gaussian. We can denote this as:

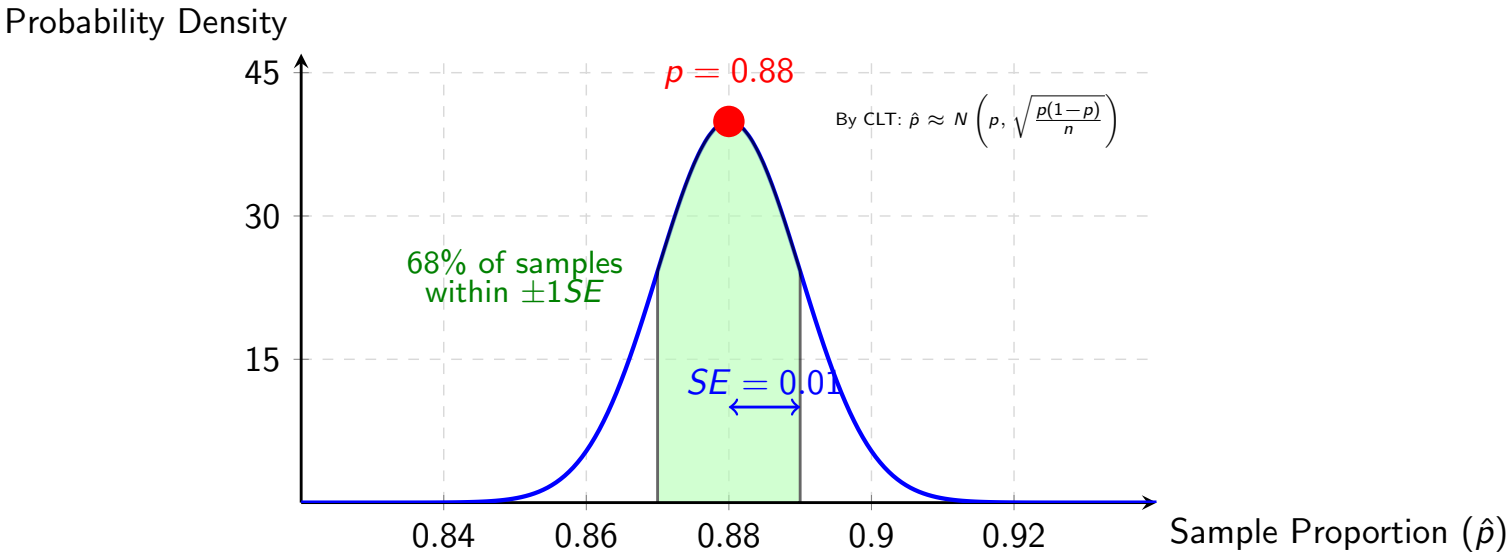
$$\hat{p} \sim \mathcal{N}\left(p, \frac{\sigma^2}{n}\right) \text{ OR } \mathcal{N}\left(\mu_p, \frac{\sigma_p^2}{n}\right) \tag{14}$$

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(b) $\mu_{\hat{p}}$ denotes the mean estimate of p, which is 0.88 (according to the CLT, the mean of the sample is the population mean if n is large).
$SE_{\hat{p}}$ denotes the sampling error, which is the the square root of the variance of the sample mean: $\sqrt{\sigma^2/n}$. Given that the sample is governed by the Binomial distribution with $\sigma^2 = p(1-p)$. Thus:

$$SE_{\hat{p}}^2 = \frac{\sigma^2}{n} = \frac{p(1-p)}{n} = \frac{0.88(0.12)}{1000}$$

$$SE_{\hat{p}} = \sqrt{\frac{0.88(0.12)}{1000}} = \boxed{0.01}$$

CLT application: sample proportion (cont.)

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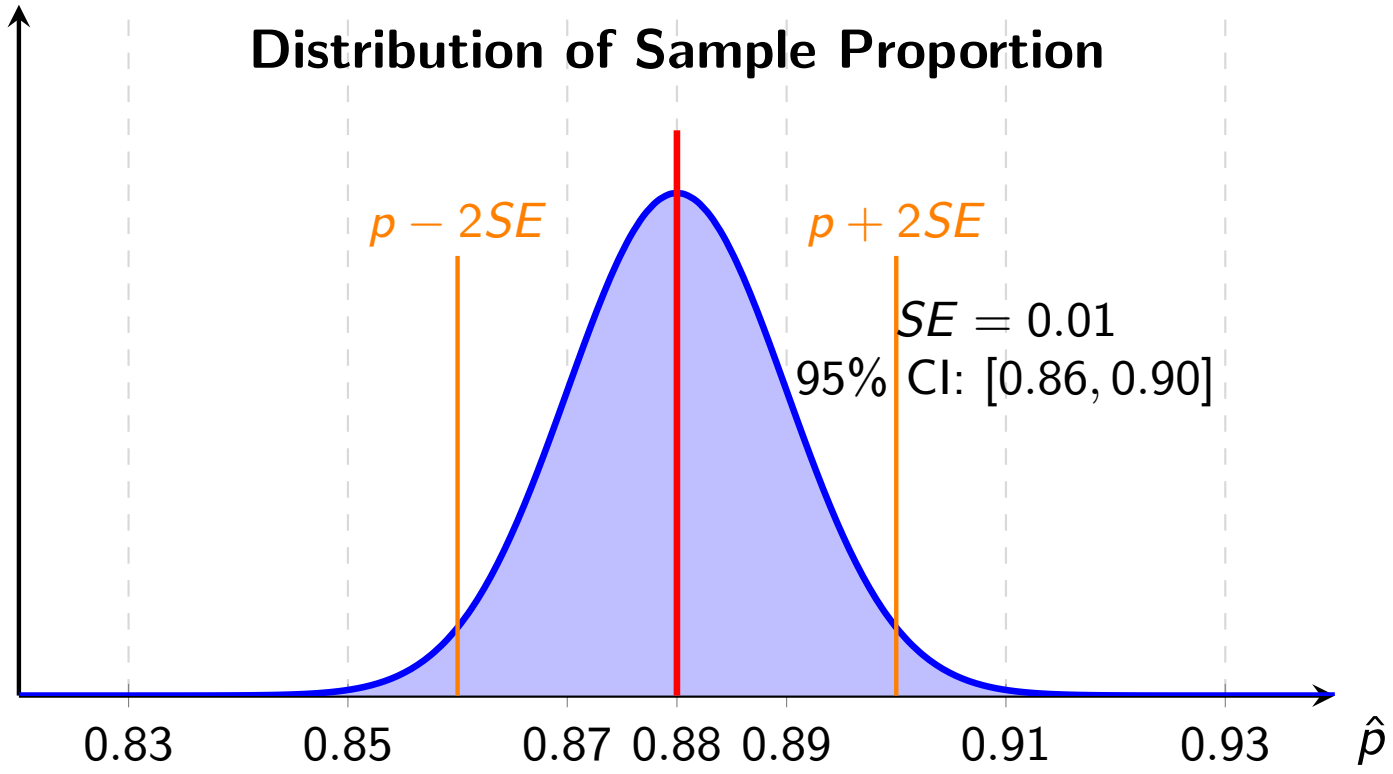


Figure: Sample proportion distribution: most samples fall within $\pm 2SE$ of the true proportion

Another application of the CLT

Example 3: Mean batch weight

A certain brand of cement is shipped in batches of 40 bags. Previous records indicate the weight of a randomly selected bag of this brand has a mean of 2.5 kg and an SD of 0.1 kg. The exact distribution is unknown.

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(a) What is the mean weight of one batch of this brand of cement?

(b) If the shipping company charges an overweight fee if a batch exceeds the mean batch weight by more than 1 kg, what is the probability that a batch will be charged?

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$$\mu_B = 40 \times 2.5$$

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Summary

- Desired properties of point estimates: unbiasedness and efficiency
- Distribution of sample proportions (or other parameters) is called a sampling distribution
- When n is sufficiently large and observations are independent, the sample proportion (or other parameter) follows a normal distribution
- The success-failure condition can be used to determine if n is large enough for the CLT to hold (for a sample proportion)

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